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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Load the image from the Colab environment
img = cv2.imread('image1.jpg')

if img is not None:
    # Laplacian mask with diagonal neighbors extension
    # The center is -8 to account for all 8 surrounding neighbors (4 sides
    kernel = np.array([[1, 1, 1],
                        [1, -8, 1],
                        [1, 1, 1]])

    # Apply the filter to extract details
    details = cv2.filter2D(img, -1, kernel)

    # Sharpen by subtracting the Laplacian (details) from the original
    sharpened = cv2.subtract(img, details)

    # Save and display
    cv2.imwrite('Sharpened_Image_Diagonal.jpg', sharpened)
    print("Sharpened Image (Laplacian with Diagonal Extension):")
```

```
cv2_imshow(sharpened)  
else:  
    print("Error: 'image1.jpg' not found. Please ensure the image is uploaded")
```

Sharpened Image (Laplacian with Diagonal Extension):

